Board A Robot I/O hub

STM32F407VGT6, LQFP-100, 82 GPIOs

Name _____

Date _____

① What must the board do?

Write each need in plain words, then the peripheral it calls for.

② Pin budget

Interface	Pins	Interface	Pins

Total _____ **of 82 GPIOs** **Spare pins:** _____

③ Parts list

Role	What to look for	Chosen part	LCSC #	JLCPCB
Microcontroller	STM32F407VGT6, LQFP-100	**STM32F407VGT6**		☐ Basic ☐ Ext.
3.3 V regulator	Buck if fed from 12 V; MCU + PHY + sensors, with margin			☐ Basic ☐ Ext.
HSE crystal	8 or 25 MHz, low ESR, gm_{crit} under 1 mA/V			☐ Basic ☐ Ext.
HSE load caps ×2	$2 \times (C_L - C_{stray})$, C0G			☐ Basic ☐ Ext.
LSE crystal	32.768 kHz, about 6 pF C_L, gm_{crit} under 0.56 µA/V			☐ Basic ☐ Ext.
VCAP caps	2 × 2.2 µF ceramic, ESR under 2 Ω			☐ Basic ☐ Ext.
Decoupling	100 nF per VDD pin + 4.7–10 µF; VDDA 100 nF + 1 µF			☐ Basic ☐ Ext.
Ethernet PHY	RMII, 3.3 V; where does its clock come from?			☐ Basic ☐ Ext.
RJ45 jack	Built-in magnetics			☐ Basic ☐ Ext.
CAN transceiver ×2	3.3 V supply			☐ Basic ☐ Ext.
USB connector + ESD	Connector plus ESD array			☐ Basic ☐ Ext.
Debug header	SWD: SWDIO, SWCLK, NRST, 3.3 V, GND			☐ Basic ☐ Ext.
				☐ Basic ☐ Ext.
				☐ Basic ☐ Ext.

Board B Fast control loop

STM32F446RET6, LQFP-64, 50 GPIOs

Name _____

Date _____

1 What must the board do?

Write each need in plain words, then the peripheral it calls for.

2 Pin budget

Interface	Pins	Interface	Pins

Total _____ **of 50 GPIOs** **Spare pins:** _____

3 Parts list

Role	What to look for	Chosen part	LCSC #	JLCPCB
Microcontroller	STM32F446RET6, LQFP-64 (Extended at JLCPCB)	**STM32F446RET6**		☐ Basic ☐ Ext.
3.3 V regulator	Buck from the flight or motor battery			☐ Basic ☐ Ext.
HSE crystal	8 MHz; PLLSAI makes USB's 48 MHz at 180 MHz			☐ Basic ☐ Ext.
HSE load caps ×2	$2 \times (C_L - C_{stray})$, C0G			☐ Basic ☐ Ext.
LSE crystal	Only if the RTC is used; low ESR, about 6 pF C_L			☐ Basic ☐ Ext.
VCAP cap	1 × 4.7 µF ceramic, ESR under 1 Ω			☐ Basic ☐ Ext.
Decoupling	100 nF per VDD pin + 4.7–10 µF; VDDA 100 nF + 1 µF			☐ Basic ☐ Ext.
Quad-SPI flash	3.3 V NOR, quad I/O, sized for logs			☐ Basic ☐ Ext.
Audio codec / DAC	I^2S/SAI input, the sample rates you need			☐ Basic ☐ Ext.
Gate driver	Accepts 3.3 V logic inputs			☐ Basic ☐ Ext.
USB-C + ESD	Connector plus ESD array			☐ Basic ☐ Ext.
Debug header	SWD: SWDIO, SWCLK, NRST, 3.3 V, GND			☐ Basic ☐ Ext.
				☐ Basic ☐ Ext.
				☐ Basic ☐ Ext.

Crystal worksheet

One column for the system clock (HSE), one for the RTC (LSE)

Name _____

Date _____

Pick the HSE frequency

Board _____ Core clock _____ Crystal _____

- ☐ Crystal ÷ M gives 2 MHz (ST's recommended PLL input)
- ☐ The PLL reaches the core clock exactly
- ☐ 48 MHz for USB, from PLL Q or (F446) PLLSAI
- ☐ F446 above 168 MHz: over-drive mode enabled

HSE: from the crystal's datasheet

Frequency	_____	MHz
ESR, max	_____	Ω
C_0, shunt	_____	pF
C_L, load	_____	pF
Tolerance / stability	_____	ppm
Drive level, max	_____	µW

LSE: from the crystal's datasheet

Frequency	**32.768**	kHz
ESR, max	_____	Ω
C_0, shunt	_____	pF
C_L, load	_____	pF
Tolerance / stability	_____	ppm
Drive level, max	_____	µW

Will it start?

$$gm_{crit} = 4 \times ESR \times (2\pi F)^2 \times (C_0 + C_L)^2$$

gm_{crit} = _____ mA/V

Limit for the F407 and F446 HSE: **1 mA/V**.

- ☐ Under the limit (gain margin of 5 or more)

Load capacitors

$$C1 = C2 = 2 \times (C_L - C_{stray})$$

2 × (	_____	pF −
C_{stray}	_____	pF)
= each cap	_____	pF
Nearest standard	_____	pF, C0G

Example: 8 MHz, ESR 80 Ω, C_0 3 pF, C_L 20 pF gives 0.43 mA/V (passes). With C_{stray} 5 pF, each cap is 30 pF.

Will it start?

$$gm_{crit} = 4 \times ESR \times (2\pi F)^2 \times (C_0 + C_L)^2$$

gm_{crit} = _____ µA/V

Limits: F407 **0.56 µA/V**. F446 **0.56 µA/V** (low drive) or **1.5 µA/V** (high drive).

- ☐ Under the limit (gain margin of 5 or more)

Load capacitors

$$C1 = C2 = 2 \times (C_L - C_{stray})$$

2 × (	_____	pF −
C_{stray}	_____	pF)
= each cap	_____	pF
Nearest standard	_____	pF, C0G

Example: a common 12.5 pF watch crystal (ESR 70 kΩ, C_0 1 pF) gives 2.16 µA/V and fails. A 6 pF, 50 kΩ crystal gives 0.42 µA/V and passes.

Layout checklist

- ☐ Crystal and both caps next to OSC_IN / OSC_OUT
- ☐ Short, roughly equal traces with no vias
- ☐ Caps grounded together near the MCU's VSS
- ☐ Ground guard ring; nothing routed underneath
- ☐ Away from motor drivers and switching regulators
- ☐ Checked in CubeMX's clock tab